TEXORA · COURSE NOTES · LECTURE SHEET

Acids and Bases

Three definitions of widening power, the pH scale and its arithmetic, conjugate pairs, buffers, indicators and the hydrolysis of salts.

CHE-1105 · CHEMISTRY-I · TOPIC 3 · LEVEL 1 TERM 1

1. Three definitions, each wider than the last

THEORY	ACID IS…	BASE IS… · AND THE LIMITS
Arrhenius	A substance producing H^+ in water ($HCl \rightarrow H^+ + Cl^-$)	A substance producing OH^- in water (NaOH). Fails outside water and for bases like NH_3 that contain no OH.
Brønsted–Lowry	A proton (H^+) donor	A proton acceptor. Works in any solvent; creates the idea of conjugate pairs. $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$ now reads perfectly: water is the acid here.
Lewis	An electron-pair acceptor (BF_3, H^+, metal ions)	An electron-pair donor (NH_3, OH^-, H_2O). The widest: includes reactions with no protons at all, such as $BF_3 + NH_3$.

Every Arrhenius acid is a Brønsted acid and every Brønsted acid is a Lewis acid — the definitions nest. Examination answers should state which theory is in use; industrial practice (dye baths, tanneries) speaks Brønsted–Lowry through the pH meter, while coordination chemistry speaks Lewis.

2. Conjugate pairs and strength

In a Brønsted reaction the acid, having donated its proton, becomes a **conjugate base**; the base, having accepted one, becomes a **conjugate acid**: $HA + H_2O \rightleftharpoons H_3O^+ + A^-$, with pairs HA/A^- and H_3O^+/H_2O. The rule that organises all of acid–base behaviour: **the stronger the acid, the weaker its conjugate base** (and vice versa) — a strong acid's anion has almost no appetite for protons. **Amphiprotic** species (H_2O, HCO_3^-, HSO_4^-) can play either role. Acid strength in water is measured by the equilibrium constant K_a for $HA + H_2O \rightleftharpoons H_3O^+ + A^-$, quoted as $pK_a = -\log K_a$: the smaller the pK_a, the stronger the acid. Strong acids (HCl, H_2SO_4 first proton, HNO_3) dissociate essentially completely; weak acids

(acetic acid, $pK_a \approx 4.74$) dissociate a few percent at ordinary concentrations. The levelling effect of water caps measurable strength at H_3O^+ — acids stronger than H_3O^+ all appear equally strong in water.

3. The pH scale and its arithmetic

Water itself ionises slightly: $H_2O + H_2O \rightleftharpoons H_3O^+ + OH^-$, with $K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}$ at 25 °C, so pure water has $[H_3O^+] = [OH^-] = 10^{-7}$ mol L^{-1}. The pH scale compresses this range:

$$pH = -\log_{10} [H_3O^+], \quad pH + pOH = 14 \text{ (at 25 °C)}.$$

WORKED EXAMPLES — THREE QUICK CALCULATIONS

(a) Strong acid: 0.01 mol L^{-1} HCl is fully dissociated, $[H_3O^+] = 10^{-2}$, pH = **2**. **(b) Strong base:** 0.001 mol L^{-1} NaOH gives $[OH^-] = 10^{-3}$, pOH = 3, pH = **11**. **(c) Weak acid:** 0.1 mol L^{-1} acetic acid ($K_a = 1.8 \times 10^{-5}$): for a weak acid $[H_3O^+] \approx \sqrt{(K_a \cdot c)} = \sqrt{(1.8 \times 10^{-6})} \approx 1.34 \times 10^{-3}$, so pH ≈ **2.87** — a hundred times less acidic than the same concentration of HCl, which is the whole meaning of “weak”.

4. Buffers

A **buffer solution** — a weak acid with its conjugate base (or a weak base with its conjugate acid) — resists pH change when small amounts of acid or alkali are added: added H^+ is mopped up by the base partner, added OH^- by the acid partner, and the equilibrium shifts instead of the pH. The buffer’s pH is given by the **Henderson–Hasselbalch equation**:

$$pH = pK_a + \log ([\text{conjugate base}] / [\text{acid}]).$$

An acetic acid/acetate buffer with equal concentrations sits at $pH = pK_a = 4.74$; ten times more acetate than acid raises it exactly one unit, to 5.74. Buffers hold the pH of dye baths, fermentation broths and blood (bicarbonate buffer, pH 7.4) against everything the process throws at them — a buffer is chosen with pK_a near the target pH, and its capacity is greatest at equal concentrations.

5. Indicators and titrations

An **acid–base indicator** is itself a weak acid whose two forms (HIn and In^-) have different colours; it changes colour over a range of about $pK_{In} \pm 1$. Choosing an indicator means matching its range to the steep part of the titration curve: methyl orange (about 3.1–4.4) for strong acid–weak base, phenolphthalein (about 8.2–10.0) for weak acid–strong base, and almost anything for strong–strong titrations, whose curve leaps through the whole middle range. The equivalence point pH is 7 only when

both partners are strong; a weak acid titrated with strong base ends alkaline (its anion hydrolyses — next section), which is precisely why phenolphthalein, not methyl orange, belongs there.

6. Hydrolysis of salts

Dissolved salts can make water acidic or alkaline when their ions react with it — **hydrolysis**:

SALT TYPE	EXAMPLE	SOLUTION
Strong acid + strong base	NaCl	Neutral — neither ion hydrolyses.
Strong acid + weak base	NH_4Cl	Acidic — NH_4^+ donates protons to water.
Weak acid + strong base	CH_3COONa	Alkaline — CH_3COO^- accepts protons from water, releasing OH^-.
Weak acid + weak base	CH_3COONH_4	Near neutral — both hydrolyse; the balance is set by K_a vs K_b.

7. The textile thread

No topic of this course is closer to the dyehouse. Scouring runs on hot alkali (which cellulose tolerates and wool does not); disperse and reactive dyeing are pH-programmed, with buffers holding the bath while alkali is dosed; acid, left drying on cotton, tenders it — the pH arithmetic of §3 is the difference between a level dyeing and a rejected lot. An engineer who cannot compute a buffer is not yet safe in a wet-processing plant.

8. Glossary — key terms of this topic

TERM	MEANING
Conjugate pair	An acid and the base formed when it loses a proton (HA/A^-).
Amphiprotic	Able to donate or accept a proton depending on the partner (H_2O, HCO_3^-).
K_w	The ionic product of water, $[H_3O^+][OH^-] = 10^{-14}$ at 25 °C.
Buffer	A mixture of a weak acid and its conjugate base (or vice versa) resisting pH change.
Equivalence point	The point in a titration where acid and base have been mixed in exact stoichiometric amounts.
Hydrolysis	Reaction of a salt’s ion with water, producing an acidic or alkaline solution.

9. Key points and summary

- Arrhenius → Brønsted–Lowry → Lewis: each definition widens the last; conjugate pairs and "strong acid, weak conjugate base" organise the Brønsted world.
- pH = −log $[H_3O^+]$; strong acids dissociate fully, weak acids follow $[H_3O^+] \approx \sqrt{(K_a c)}$; pH + pOH = 14 at 25 °C.
- Buffers obey Henderson–Hasselbalch and are strongest at equal concentrations; indicators must be matched to the titration curve's steep section; salts hydrolyse acidic, alkaline or neutral by parentage.

10. Practice

Answer in your own words; brief model answers follow.

1. Classify NH_3 as a base under all three theories, noting where Arrhenius strains.
2. Write the conjugate base of HCO_3^- and the conjugate acid of HCO_3^- (it is amphiprotic — show both).
3. Calculate the pH of (a) 0.05 mol L^{-1} HCl, (b) 0.02 mol L^{-1} NaOH.
4. A buffer contains 0.20 mol L^{-1} acetic acid and 0.20 mol L^{-1} sodium acetate (pK_a = 4.74). Find its pH, then the pH if the acetate is raised to 2.0 mol L^{-1}.
5. Predict whether solutions of (a) KNO_3, (b) Na_2CO_3, (c) NH_4NO_3 are acidic, neutral or alkaline, with the hydrolysis reason.
6. A weak acid is titrated with NaOH. Choose between methyl orange (3.1–4.4) and phenolphthalein (8.2–10.0) and justify.
7. BF_3 + NH_3 → $F_3B–NH_3$. Identify the Lewis acid and base, and why Brønsted–Lowry cannot describe this reaction.
8. Two acids have pK_a values 3.2 and 9.1. Which is stronger, and how many times larger is its K_a?

Answer key

1. Brønsted: accepts a proton from water. Lewis: donates its lone pair. Arrhenius: it produces OH^- only indirectly by reacting with water — the strain that Brønsted fixed.
2. Conjugate base: CO_3^{2-}. Conjugate acid: H_2CO_3.
3. (a) $[H_3O^+]$ = 0.05, pH = −log 0.05 ≈ 1.30. (b) pOH = −log 0.02 ≈ 1.70, pH = 14 − 1.70 = 12.30.
4. Equal concentrations: pH = 4.74. Acetate × 10: pH = 4.74 + log 10 = 5.74.
5. (a) Neutral (strong acid + strong base parentage). (b) Alkaline — CO_3^{2-} hydrolyses, accepting protons from water. (c) Acidic — NH_4^+ hydrolyses, donating protons.

6. Phenolphthalein — the equivalence point of a weak acid–strong base titration is alkaline (the anion hydrolyses), inside phenolphthalein's range, far above methyl orange's.
7. Lewis acid: BF_3 (accepts the pair into its empty orbital). Lewis base: NH_3 (donates the pair). No proton is transferred, so Brønsted–Lowry has nothing to describe.
8. pK_a 3.2 is stronger. K_a ratio = 10^(9.1 − 3.2) = 10^5.9 ≈ 8×10^5 times larger.
